

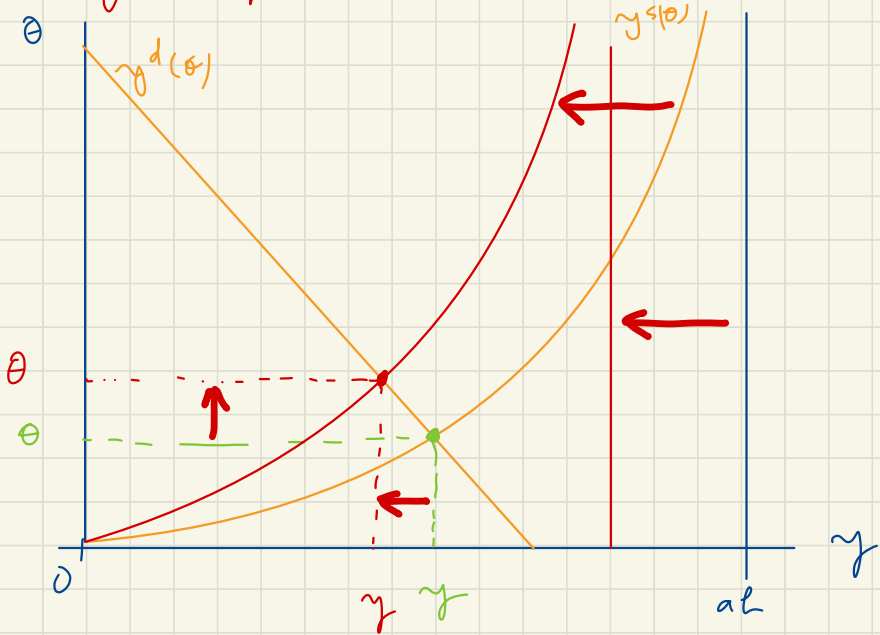
Aggregate Supply Shocks with Fixed Inflation

Pascal Michailat
<https://pascalmichailat.org/w/>

Negative AS shock:

$$y^s(\theta) = \frac{f(\theta)}{1+f(\theta)} \cdot a \cdot \underline{h}$$

Lower labor-force participation: lower h



After reduction in labor force h

tightness:

output

employment

unemployment rate

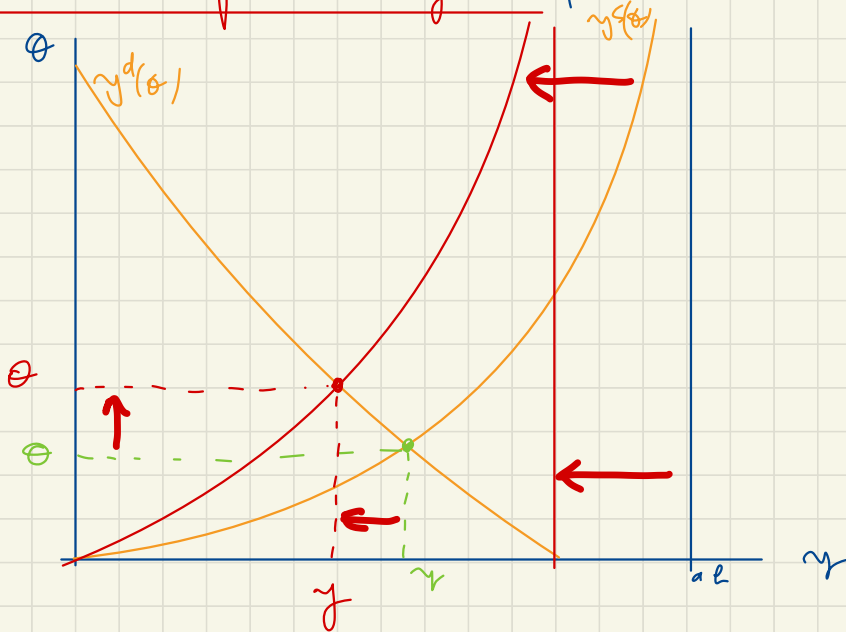
θ	$\uparrow$
y	$\downarrow$

different from AD shocks

$$l = y/a \quad \text{so} \quad l \downarrow$$

$$u = \frac{1}{1+f(\theta)} \quad \text{so} \quad u \downarrow$$

Lower labor productivity a (another AS shock)



Lower labor productivity a

Tightness

a ↑

Output

y ↓

Unemployment rate

$$u = \frac{1}{1 + f(\theta)} \text{ so } u \downarrow$$

Employment

$$\ell = (1 - u) \times h$$

so

$\ell \uparrow$

specific to productivity shocks

Evidence that $\ell \downarrow$ when $a \uparrow$

Baun et al (2006)